

Given the following table.
calculate the median, mean and mode.

Class	[6-12]	[12-18]	[18-24]	[24-30]	[30-36]	[36-42]
n_i	4	7	10	15	8	6
x_i	9	15	21	27	33	39
I.C.F	4	11	21	36	44	50

Solution:
calculate x_i and I.C.F, $N = 50$

$$\text{mean} = \frac{\sum x_i n_i}{N} = \frac{1154}{50} = 23.28$$

$$\text{median} = L + \frac{\frac{N}{2} - n_{i-1}}{n_i} \times h \Rightarrow \boxed{\text{Class: } [24-30]}$$

$$\text{median} = 24 + \frac{25 - 22}{15} \times 6 = 25.6$$

$$\text{mode} = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times h$$

$$\text{mode} = 24 + \frac{15 - 10}{5 + 7} \times 6 = 26.5$$